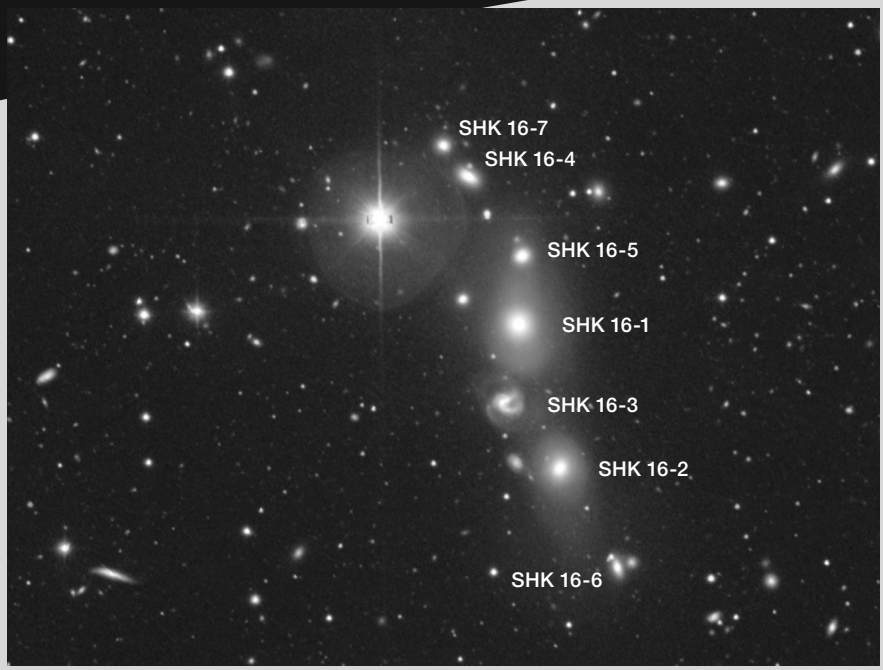

◀ **SHAKHBAZIAN 16** You'll want to get your bigger telescopes out and make sure you have ideal observing conditions to snag the targets outlined here, including SHK 16.

The Shakhbazian Challenge

These compact galaxy groups will test your observing skills and equipment.

Many deep-sky enthusiasts are fans of Canadian astronomer Paul Hickson's list of compact groups of galaxies (HCG), which includes Stephan's Quintet (*S&T:* Sept. 2024, p. 28). Count me in the club — my first article in the March 1999 issue of *S&T* is titled "Quintets, Sextets, and Septets: Exploring Hickson Compact Groups." However, Hickson's 1982 catalog was neither the first nor the largest compilation of compact groups of galaxies. In the 1970s, astronomers at the Byurakan Astronomical Observatory (BAO) in Armenia conducted a lesser-known but more extensive survey.

The story began in 1957 when BAO astronomer Romela Karapet Shakhbazian (also spelled Shahbazian) discovered a small group of a dozen faint red objects in southern Ursa Major. Based on their stellar appearance on plates in the *National Geographic Society – Palomar Observatory Sky Survey* (*POSS*), she thought it was a cluster of faint stars at a distance of 425,000 light-years. The following year, American-Canadian astronomer Helen Sawyer Hogg announced the globular cluster candidate "Shak" in the *Journal of the Royal Astronomical Society of Canada*. But deeper images taken in 1962 with Lick Observatory's 120-inch reflector suggested the group was an accidental assembly of stars and background galaxies rather than a star cluster.

That's where things stood until 1973 when Lick astronomers Lloyd Robinson and Joseph Wampler captured the spectra of five group members. These galaxies were all luminous red ellipticals with a redshift of $z = 0.117$, indicating a distance of 1.6 billion light-years! With 17 galaxies squeezed into a 1′ circle, Robinson and Wampler concluded, "Shakhbazian I is one of the densest galaxy clusters known."

This study prompted a systematic search at the BAO for similar groups on more than 200 *POSS* prints. An Armenian and German collaboration published 10 papers between 1973 and 1979. They listed 377 compact groups using several criteria involving, among others: number of members (five to 15); magnitude (between 14th and 19th on red *POSS* images); and that all galaxies should be extremely red with a high surface brightness and no diffuse border.

Later photometric and spectroscopic studies established the vast majority are gravitationally bound groups (several in chainlike configurations) and not chance alignments. However, due to the limited resolution of the *POSS* images, some of the original members turned out to be faint Milky Way stars. Furthermore, the Shakhbazian groups (SHK) encompass a range of spatial densities, with many individual galaxies being standard ellipticals.

In comparison to HCGs, the SHK groups are dimmer, more distant, and richer in early-type galaxies (E and S0). To snag more than a few members, I suggest using an 18-inch or larger scope in preferably Bortle-2 skies and good transparency. Employing high magnifications enhances the visibility of the nearly stellar members. Patience is also crucial — each group contained phantom glows that teetered on the edge of perception in my 18-inch and 24-inch scopes.

Chains in Draco and Ursa Minor

Let's start with a few of the more accessible groups. **SHK 16** is a remarkable north-south chain of compact galaxies in southern Draco. Before Shakhbazian cataloged the chain in 1973, Russian astronomer Boris Vorontsov-Vel'yaminov included several individual members in the 1962 *Morphological Catalogue of Galaxies, Part 1*. Halton Arp added the group (Arp 330) to his 1966 "*Atlas of Peculiar Galaxies*" and described, "Five galaxies in chain quite compact; 6th of low surface brightness." The latter object is a blue spiral showing evidence of tidal distortion.

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To find SHK 16, hop 2° east-northeast of 16 and 17 Draconis, an easy 5.4-magnitude binocular pair. The galaxy chain spans 5′ in length and passes just west of the 9.2-magnitude star HD 234352. Once you've identified the field, I'd recommend using at least 250× to spread the members away from this stellar signpost. I've observed this delightful chain at least a half-dozen times over the years. My favorite view, though, was through my 18-inch from the 8,500-foot-elevation trailhead to Lassen Peak in Northern California.

Using 328×, I spotted **SHK 16-1** just 2′ southwest of the bright star. Despite a magnitude of 14.6, the galaxy has a compact diameter of 18″ and a relatively high surface brightness. Two fainter companions, **SHK 16-5** and **SHK 16-3**, bracket the galaxy north and south by 0.8′ and resemble bloated stars. **SHK 16-2** marks the southern end of the chain, and I could hold it steady. By scanning 1′ northwest of the star, I caught **SHK 16-4**, a droplet of light near the glare. I also noticed PGC 2817454 just 3′ north of SHK 16-4. It displayed a 2:1 elongation and was comparable in visibility to SHK 16-1 and SHK 16-2.

SHK 166 is a similar 8′ linear chain, conveniently located only ½° south-southeast of 4.2-magnitude Epsilon (ε) Ursae Minoris, one of the handle stars of the Little Dipper. The group comprises several featureless elliptical and lenticular galaxies queued up from southwest to northeast.

At 281×, I first spied **SHK 166-1**, an easy 20″ fuzzball just 50″ northeast of a 13th-magnitude star. I find this star helpful to orient the group members. Three galaxies, **SHK 166-4**, **SHK 166-3**, and **SHK 166-2**, line up to the southwest of SHK 166-1. The closest is SHK 166-4, an intermittent glow just 40″ southwest of the star, while SHK 166-3 is comparable in brightness to SHK 166-1. Two feeble smudges lie close to the northeast of SHK 166-1: **SHK 166-7** is a miniature oval angling north-south with the terribly elusive **SHK 166-8** to its east. By taking my time and focusing on each member individually, I managed to tally seven galaxies including PGC 59182, a 12″ speck about 5′ south of the chain. Although not assigned an SHK designation, it has the same redshift as the chain.

Armenian Astronomer

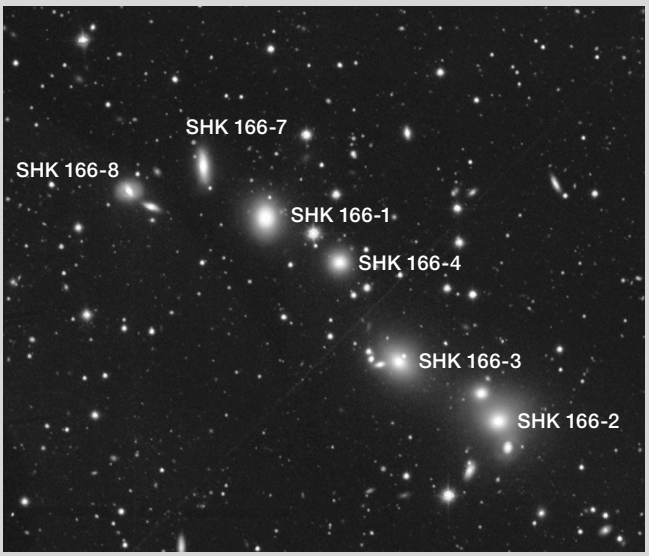
Romela Shakhbazian was born in 1925 in the southern Russian city of Rostov-on-Don. She was a member of the BAO staff from 1953 until her retirement in 1995. Shakhbazian defended her PhD thesis in 1970 under the supervision of Viktor Ambartsumian, the observatory's founder and one of the leading astrophysicists of the 20th century. As well as compact galaxy groups, she also researched stellar associations in nearby galaxies and extragalactic supernovae. Shakhbazian passed away in 2012 at the age of 87.

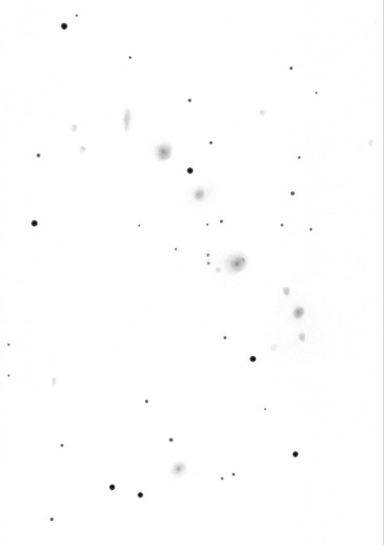
Pileups in Coma and Leo

SHK 202 is one of the nearer and brighter of the Shakhbazian groups, but there's a spoiler. The members huddle near the 6.6-magnitude star 10 Comae Berenices, whose brilliance may overwhelm their anemic glows.

Before exploring SHK 202, let's take a peek at NGC 4251, a 10.7-magnitude barred lenticular located ½° southwest of 10 Com. It spans 3.6′ × 1.5′ from east to west and features a bright, bulging core and a star-like nucleus. Shift 18′ to the northeast, and you'll arrive at 14th-magnitude IC 777. Through my 18-inch, this foreground galaxy tilts 2:1 northwest to southeast and outshines the nearby members of SHK 202.

Since SHK 202-1 lies a mere 5′ south of 10 Com, I used 390× and nudged the stellar nuisance outside the field of view. SHK 202-1 appeared as a 20″ × 15″ oval patch with a decent surface brightness. The similarly fuzzy SHK 202-3 floats just 2′ north. If you're successful with this pair, SHK 202-5 is a tiny toughie squeezed in between! SHK 202-6 is the lone member on the northwest side of 10 Com, while SHK 202-2 is 6′ southeast of the star. Both match SHK 202-1 in size and brightness. Want one more challenge? I barely glimpsed SHK 202-4 in





▲ **SHAKHBAZIAN 166** The galaxies in this pleasing chain are sprinkled from northeast to southwest. German amateur Uwe Glahn sketched it using his 27-inch f/4 telescope at 419×.

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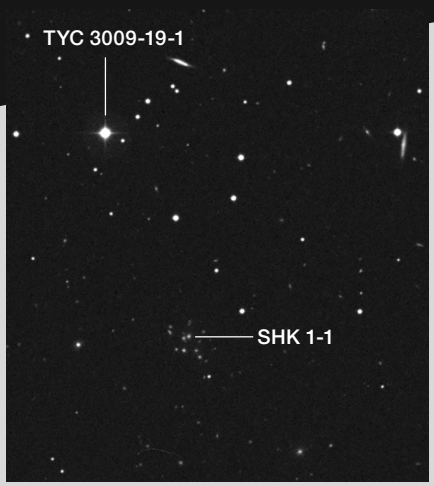

◀ **SHAKHBAZIAN 1** The first galaxy group Romela Shakhbazian identified lies between the hind paws of the Great Bear. It's the most challenging collection described here — Glahn sketched the four brightest members using his 27-inch under pristine conditions.

my 18-inch — a ghostly bead of light less than 3′ southeast of 10 Com.

Next up is **SHK 352**, which occupies a cramped 2.5′ × 1.5′ plot of celestial real estate nearly 700 million light-years away. To locate our target, point towards Tau (τ) Leonis, also known as 84 Leonis, in the southeastern corner of the constellation. Tau consists of a wide pair (1.5′ separation) of 5.1- and 7.5-magnitude stars tinted light yellow and blue. It forms a binocular "double-double" with 83 Leonis, a neighboring pair 20′ northwest of Tau and consisting of 6.6- and 7.5-magnitude stars 29″ apart. From Tau, scoot 1.7° to the west-northwest, and you can't miss 10.4-magnitude NGC 3640. SHK 352 lies just 22′ to its south-southwest.

Through my 24-inch Dob at 380×, I found a clump of four petite galaxies. SHK 352-1 appeared as a faint, gray stain measuring at most 20″ × 15″. Three smaller and dimmer spots, SHK 352-2, SHK 352-3, and SHK 352-4, adjoin its north and east. Moving 5′ south of SHK 352-1, NGC 3644 is a brighter 14th-magnitude spiral, but it's a foreground imposter at half the distance. When I examined SHK 352 through Jimi Lowrey's jumbo-size 48-inch in west Texas, four new members (numbers 5 through 8) joined the heap.

Challenges in Ursa Major

SHK 5 is better known as HCG 50 and is considered the most difficult of the 100 Hickson groups. Locating the field is easy — just head 20′ east-southeast of M97, the Owl Nebula. But SHK 5's five members are stuffed into a minuscule 40″ circle, with daunting magnitudes ranging from 17.2 to 18.7.

The key to pinpointing the group is an isosceles triangle formed by two 13th- and 15th-magnitude stars separated by 2.7′ east to west. A fainter 16th-magnitude star 2′ south completes the triangle. You'll need to spot this star easily, as the members of SHK 5 are much dimmer. Using my 24-inch scope at 200×, I navigated to the field and centered the star. I bumped the magnification up to 450× and bingo! SHK 5-1 (HCG 50A) popped into view just 20″ northeast of the dim star. The subtle, very compact glow was quite satisfying, considering the ancient light was emitted 1.9 billion years ago!

What about **SHK 1**, the prototype? Detecting any of its members requires large optics, dark and transparent skies, and an experienced eye. I've only observed the cluster through Jimi's 48-inch and McDonald Observatory's 82-inch Otto Struve Telescope. However, German amateur Uwe Glahn sketched the four brightest members in superb conditions using his 27-inch reflector at 419× to 586×. To locate this formidable target, steer 50′ west of 5th-magnitude 47 Ursae and 5′ south-southwest of a 10.3-magnitude star.

When I peered into the 48-inch at 375×, I noticed a faint, lumpy patch 1′ in diameter with a 15th-magnitude star 1.5′ to its northwest. Increasing to 488×, 17.2-magnitude **SHK 1-1** resolved, along with several other gossamer spots. The closest is 18th-magnitude SHK 1-2, barely off its east edge. Using 813×, I was amazed to see SHK 1 fracture into seven teensy galaxies down to magnitude 18.5!

If you have a large telescope and enjoy viewing galaxies in bunches, the SHK compact groups could be your next observing project. Alvin Huey provides an observing guide with finder charts and images at **www.faintfuzzies.com** to help you get started.

■ Contributing Editor **STEVE GOTTLIEB** enjoys viewing galaxy groups of all sizes from northern California. For more of his favorite observing projects, see **adventuresindeepspace.com**.

Shakhbazian Galaxy Groups

Object	Brightest Member	Mag(v)	Size	RA	Dec.
SHK 16	PGC 59049	14.6	5.2′	16^h 49.2^m	+53° 25′
SHK 166	PGC 59174	14.9	7.8′	16^h 52.8^m	+81° 38′
SHK 202	PGC 39737	14.6	9.5′	12^h 19.9^m	+28° 23′
SHK 352	PGC 34794	14.6	2.7′	11^h 21.5^m	+02° 53′
SHK 5	PGC 34447	17.2	0.7′	11^h 17.1^m	+54° 55′
SHK 1	PGC 32808	17.2	1.2′	10^h 55.1^m	+40° 27′

Angular sizes are from recent catalogs. Visually, an object's size is often smaller than the cataloged value and varies according to the aperture and magnification of the viewing instrument. Right ascension and declination are for equinox 2000.0.

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